

Dengue Outbreak Risk Prediction

Predicting subzone-level dengue cluster risk 2 weeks ahead
using Singapore government open data — NEA, LTA & SingStat

330+

SG Subzones scored

7 yrs

Real cluster data

100%

Gov open data

The business problem

01



Reactive, not proactive

NEA declares clusters only after cases accumulate. Fogging teams are deployed too late — the outbreak is already spreading.



Mosquito breeding cycle = 2 weeks

Aedes aegypti eggs hatch in 7–14 days. A 2-week advance prediction window is exactly what's needed to intervene before transmission.



The opportunity

Predict which subzones will have active clusters in the next 14 days — so NEA or pest control operators can pre-deploy fogging teams proactively.

Goal: Predict dengue cluster risk per subzone for the next 14 days — enabling proactive fogging deployment before the outbreak forms.

35,000+

dengue cases
in Singapore
2020 alone

*Largest outbreak
in recorded history*

MAS mandates
proactive pest mgmt

**\$8.6B economic
cost (WHO est.)**

ANCHOR

SGCharts Dengue Cluster Snapshots

May 2013 → Nov 2020 · 250+ bi-weekly snapshots · Community-maintained CSV

- Lat/long cluster boundaries + case counts per snapshot
- 7 years of real spatial cluster data
- Covers 2020 record outbreak (35k+ cases) — key drift event
- Treated as primary training signal: cluster present/absent per subzone per week

AUXILIARY 1

NEA Weather API

Real-time + historical · Free government API

- Rainfall, temperature, humidity by weather station
- 2-week rainfall lag = strongest dengue predictor (mosquito breeding)
- Rolling 4-week temperature mean as secondary feature

AUXILIARY 2

LTA MRT Tap Volumes + SingStat Population

LTA DataMall API · data.gov.sg

- Nearby MRT tap-in/out as neighbourhood activity proxy
- SingStat subzone population density — exposure risk weighting
- OneMap API: geocode subzones, compute spatial proximity features

LABEL (Proxy)

Synthetic Subzone Grid + Binary Label

Documented assumption — fully transparent

- 330 SG subzones gridded from URA planning boundaries
- Label: cluster present in subzone in next 14 days (1/0)
- Derived from SGCharts cluster coordinates + spatial join

✓ Follows prof's formula: real anchor (SGCharts) + real auxiliaries (NEA, LTA, SingStat) + synthetic labels only where no public data exists

Users and how they use the system

03



NEA Operations Team

When

Weekly — every Monday morning

Receives

Risk map: 330 subzones colour-coded
Low/Med/High for coming 14 days

Action

Pre-deploy fogging teams to High-risk
subzones before clusters form

Primary user — system built for their operational planning cycle



Pest Control Operators

When

Weekly — before scheduling fogging rounds

Receives

Ranked list of high-risk subzones with
confidence scores

Action

Prioritise fogging contracts, allocate teams to
predicted hotspots

Secondary user — commercial pest control companies (Rentokil, PestBusters)



Data / Ops Team

When

Ongoing — weekly monitoring

Receives

Model performance dashboard: precision,
recall, drift alerts

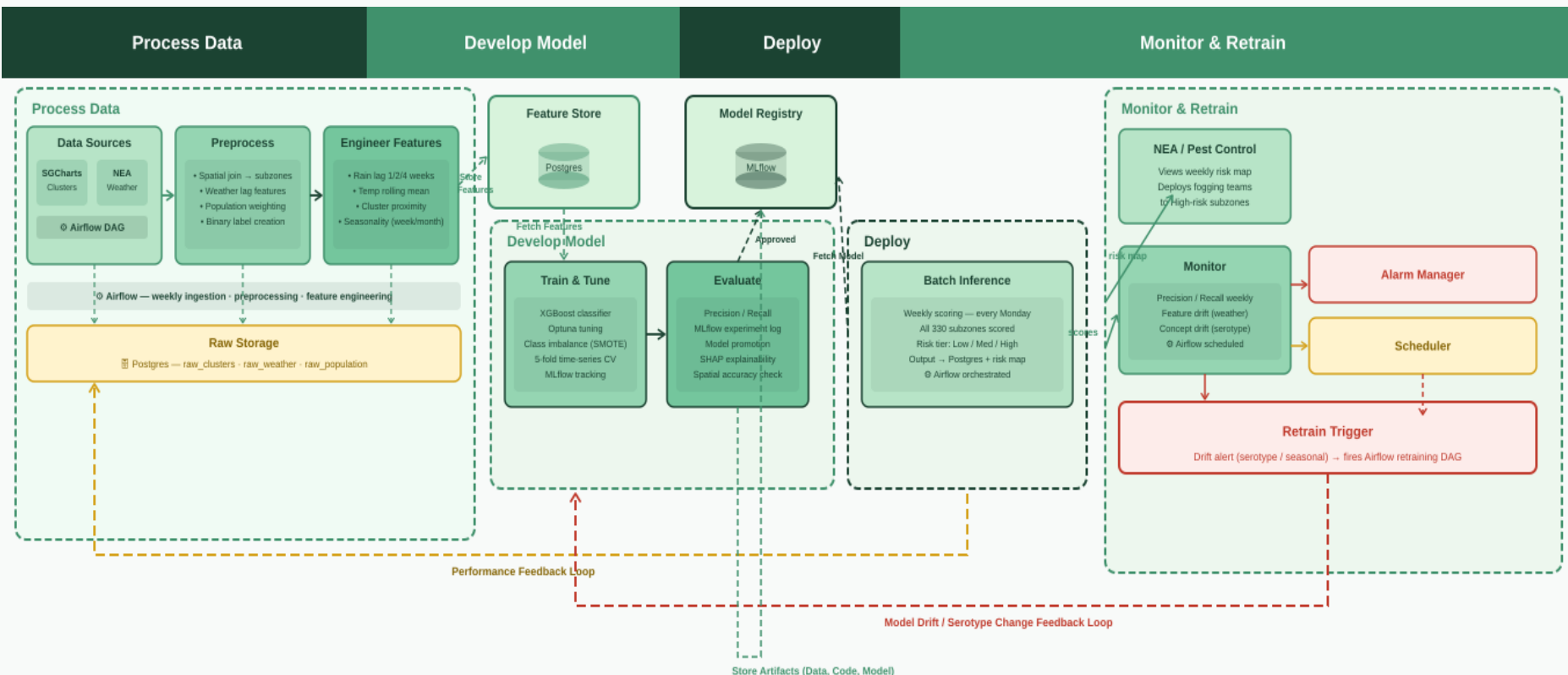
Action

Reviews weekly prediction quality; triggers
retraining when Gini drops

Internal tooling only — monitors pipeline health

High-level architecture

04



All components run locally in Docker + docker-compose — no cloud spend required

Design choices & open questions

05

Design choices — rationale

Batch over real-time inference

NEA plans fogging schedules weekly — not in milliseconds. Batch is operationally correct.

XGBoost over deep learning

Tabular spatial + temporal data; interpretable for public health reporting; faster iteration.

Time-series cross-validation

Standard k-fold leaks future data into training — temporal CV prevents data leakage.

Weekly cadence tied to Airflow

Mosquito breeding cycle is 7–14 days. Weekly prediction matches the intervention window.

Open questions — seeking feedback



SGCharts data ends 2020 — how to handle post-2020?

Option A: train on 2013–2020 and demo drift. Option B: synthesise 2021–2025 using published NEA yearly totals.



Subzone vs planning area granularity?

330 subzones is fine-grained but sparse. 55 planning areas is coarser but data-rich. Which level for prediction?



Precision vs Recall tradeoff?

False negatives (missed outbreaks) are costly. Should we optimise recall at cost of precision — over-deploying fogging teams?



Google Popular Times as auxiliary footfall signal?

Adds mobility context. But scraping ToS may be an issue. Worth pursuing or too risky for a course project?